
Chapter Two

Software Project Planning

Software Project Planning

- A Software project is concerned not only with the actual writing of software. The major activities in software project management are:
- Feasibility Study
- Planning
- Program Execution

Feasibility Study

- A feasibility study may be carried out before a project commences, Its purpose
- To establish an outline of what users require
- To ensure it is feasible to meet their needs
 - » Entirely
 - » In part
- To record initial impressions of how to meet the requirements
- To give management a feel for costs and timescales of project
- To provide as much information as possible to project manager for
 - » Estimating
 - » Purchasing and accommodation
- To help management decide whether to proceed with the project

Planning

- Probably the most time-consuming project management activity
- Continuous activity from initial concept through to system delivery.
- Plans must be regularly revised as new information becomes available
- Various different types of plan may be developed to support the main software project plan that is concerned with schedule and budget

Project planning

- Project planning is undertaken and completed before any development activity starts. Attributes for estimation
- **Project size:** What will be problem complexity in terms of the effort and time required to develop the product?
- **Cost:** How much is it going to cost to develop the project?
- **Duration:** How long is it going to take to complete development?
- **Effort:** How much effort would be required?
- **Software Project Planning:** The overall goal of project planning is to establish a pragmatic strategy for controlling, tracking, and monitoring a complex technical project.

Project Purpose and Objective

- Software project is carried out to accomplish a specific purpose, which is classified into **two** categories, namely, project objectives and business objectives.
- The commonly followed **project objectives** are listed below.
 - » Meet user requirements
 - » Meet schedule deadlines
 - » Be within budget
 - » Produce quality deliverables
- **Business objectives** related & ensure that the organizational objectives and requirements are accomplished in the project.
- Evaluate processes, Renew policies and processes, Keep the project on schedule, Improve software

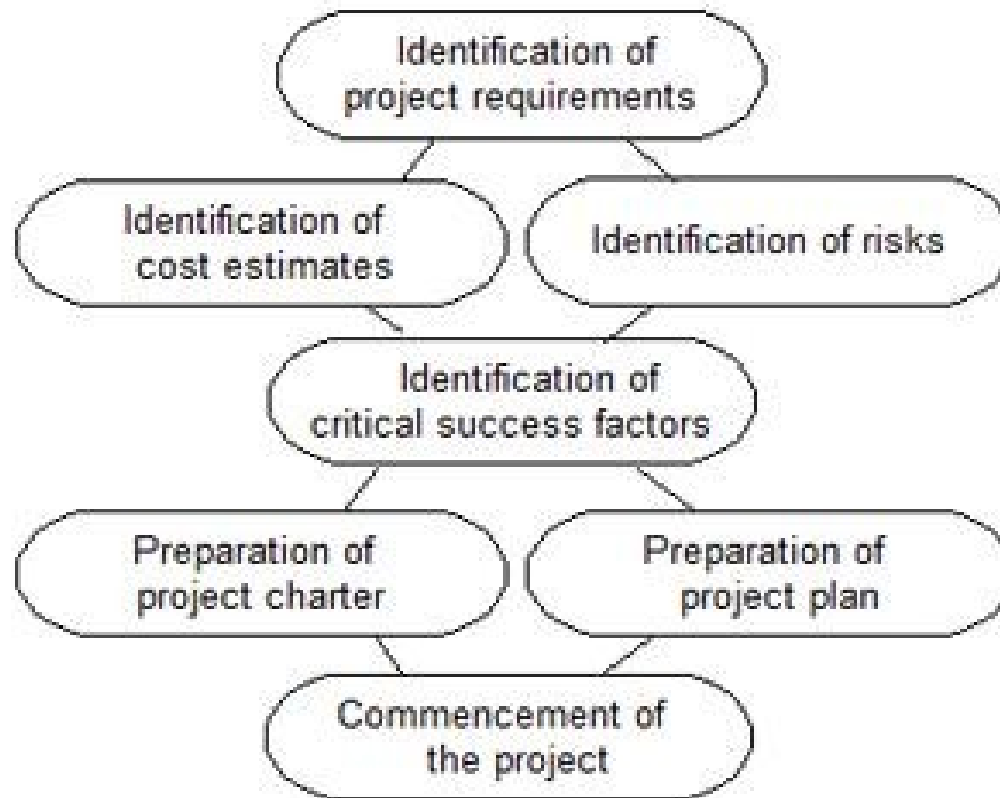
Project Scope

- Based on user requirements, the project management team determines the scope of the project before the project begins. This scope provides a detailed description of
 - » functions, features, constraints, and interfaces of the software that are to be considered.
- Project scope describes the following information.
 - » The elements included and excluded in the project
 - » The processes and entities
 - » Required functions and features
- Scope is defined using one of two techniques:
 - » A narrative description of software scope is developed after communication with all stakeholders.
 - » A set of use-cases is developed by end-users.

Project Planning Process

- The project planning process involves a set of interrelated activities followed in an orderly manner to implement user requirements followed with descriptive activities. The project planning process comprises
 - Objectives and scope of the project
 - Techniques used to perform project planning
 - Effort (in time) of individuals involved in project
 - Project schedule and milestones
 - Resources required for the project
 - Risks associated with the project.

....Project planning process comprises activities,



Project Planning Activities

Project plan development sections

- Introduction
- Project organization
- Risk analysis
- Resource requirements
- Work breakdown
- Project schedule
- Project budget breakdown

Quality Assurance Plan

- Describes the strategies and methods that are to be followed to accomplish the following objectives.
- Ensure that the project is managed, developed, and implemented in an organized way.
- Ensure that project deliverables are of acceptable quality before they are delivered to the user.

Verification and Validation (V&V) Plan

- To identify the activities that will establish compliance with the requirements (verification) and to establish that the system will meet the customers' expectations (validation).
- The verification and validation plan describes the approach, resources and schedule used for system validation.

Comprises:

- Configuration Management Plan
 - A management tool that defines, documents, controls, instructs, implements, and manages changes to various components throughout a project.
- Maintenance Plan
- Staffing Plan

Discussion

- How do you prioritize tasks in step-wise project planning and what factors do you consider when creating a software project management plan?
- What are some of the most effective software project management tools available?
 - How do we use these software?
- How do you engage proper stakeholders?

Project -I (10%)

- 1 One page summary for each group (schedule & Work BD)
 - 1 1. Introduction
 - 1 2. Project organization
 - 1 3. Risk analysis
 - 1 4. Resource requirements
 - 1 5. Work breakdown
 - 1 6. Project schedule
- 1 Every week evaluation of the progress

Project-II (%)

- Develop a project charter

Chapter Three

Work Break Down Structure (WBS) and Schedule

Introduction

- A “Project” is a set of activities which ends with specific accomplishment and which has
 - Non-routine tasks,
 - Distinct start/finish dates, and
 - Resource constraints (time/money/people/equipment).
- A project is composed of a number of related activities that are directed to the accomplishment of a desired objective.
- A project starts when at least one of its activities is ready to start.
- A project is completed when all of its activities have been completed.

Cont'd

- An activity
 - » Must have a clear start and a clear stop
 - » Must have a duration that can be forecasted
 - » **May** require the completion of other activities before it begins
 - » Should have some '**deliverables**' or **milestones** for ease of monitoring
- Scheduling: How long will it take to finish them (activities)?

Cont'd

- Root Causes for Late Software
 - » Unrealistic deadline established outside the team
 - » Changing customer requirements not reflected in schedule changes
 - » Underestimating the resources required to complete the project
 - » Risks that were not considered when project began
 - » Technical difficulties that have not been predicted in advance
 - » Human difficulties that have not been predicted in advance
 - » Miscommunication among project staff resulting in delays
 - » Failure by project management to recognize project falling behind schedule and failure to take corrective action to correct problems

Cont'd

- How to Deal With Unrealistic Schedule Demands
 - » Perform a detailed project estimate for project effort and duration using historical data.
 - » Use an **incremental** process model that will deliver critical functionality imposed by deadline, but delay other requested functionality.
 - » Meet with the customer and explain why the deadline is unrealistic using your estimates based on prior team performance.
 - » Offer an incremental development and delivery strategy as an alternative to increasing resources or allowing the schedule to slip beyond the deadline.

Cont'd

- Software Project Scheduling Principles
 - » **Compartmentalization** - the product and process must be decomposed into a manageable number of activities and tasks.
 - » **Interdependency/Crash** - tasks that can be completed in parallel must be separated from those that must be completed serially.
 - » **Time allocation** - every task has start and completion dates that take the task interdependencies into account.
 - » **Effort validation** - project manager must ensure that on any given day there are enough staff members assigned to complete the tasks within the time estimated in the project plan.

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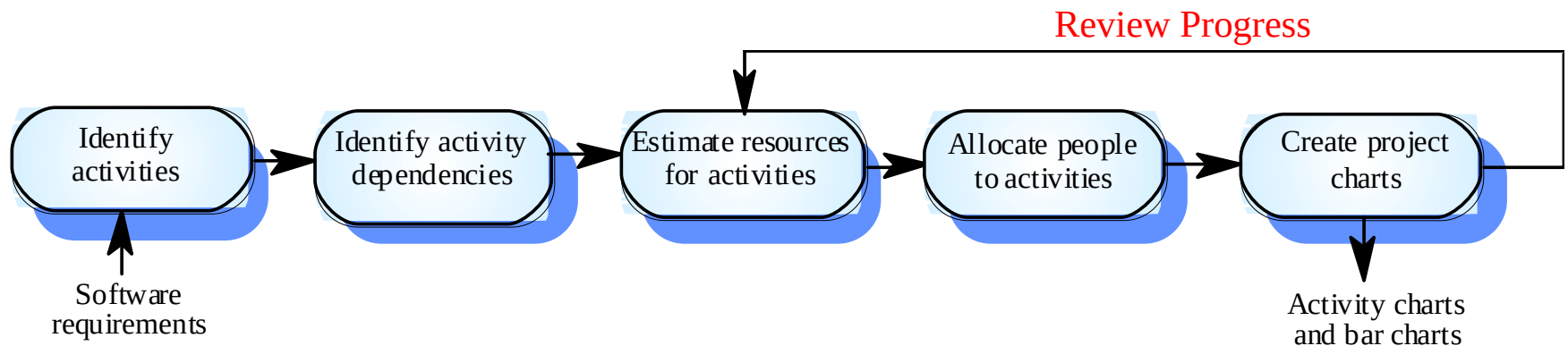
- » **Defined Responsibilities** - every scheduled task needs to be assigned to a specific team member
- » **Defined outcomes** - every task in the schedule needs to have a defined outcome (usually a work product or deliverable).
- » **Defined milestones** - a milestone is accomplished when one or more work products from an engineering task have passed quality review.

Project Scheduling Process

- Split project into *tasks* and *estimate time and resources required* to complete each.
 - » Identify the tasks that needs to be done
 - Work breakdown structure (WBS)
 - » Identify dependencies between tasks
 - Dependency Graph
 - » Estimate the duration for each task to be done
 - Schedule
- Tasks should not be too small or too large – they should last on the order of *weeks* for projects lasting *months*.

Cont'd

- Organize tasks as *concurrent* activities to make optimal use of workforce.
- *Minimize task dependencies* to avoid potential delays.
- Dependent on project managers' *intuition and experience*.



Discussion

- What are the key components of a WBS?
- How can a WBS help in managing project scope and budget?
- How do you ensure that all project tasks are included in the WBS?

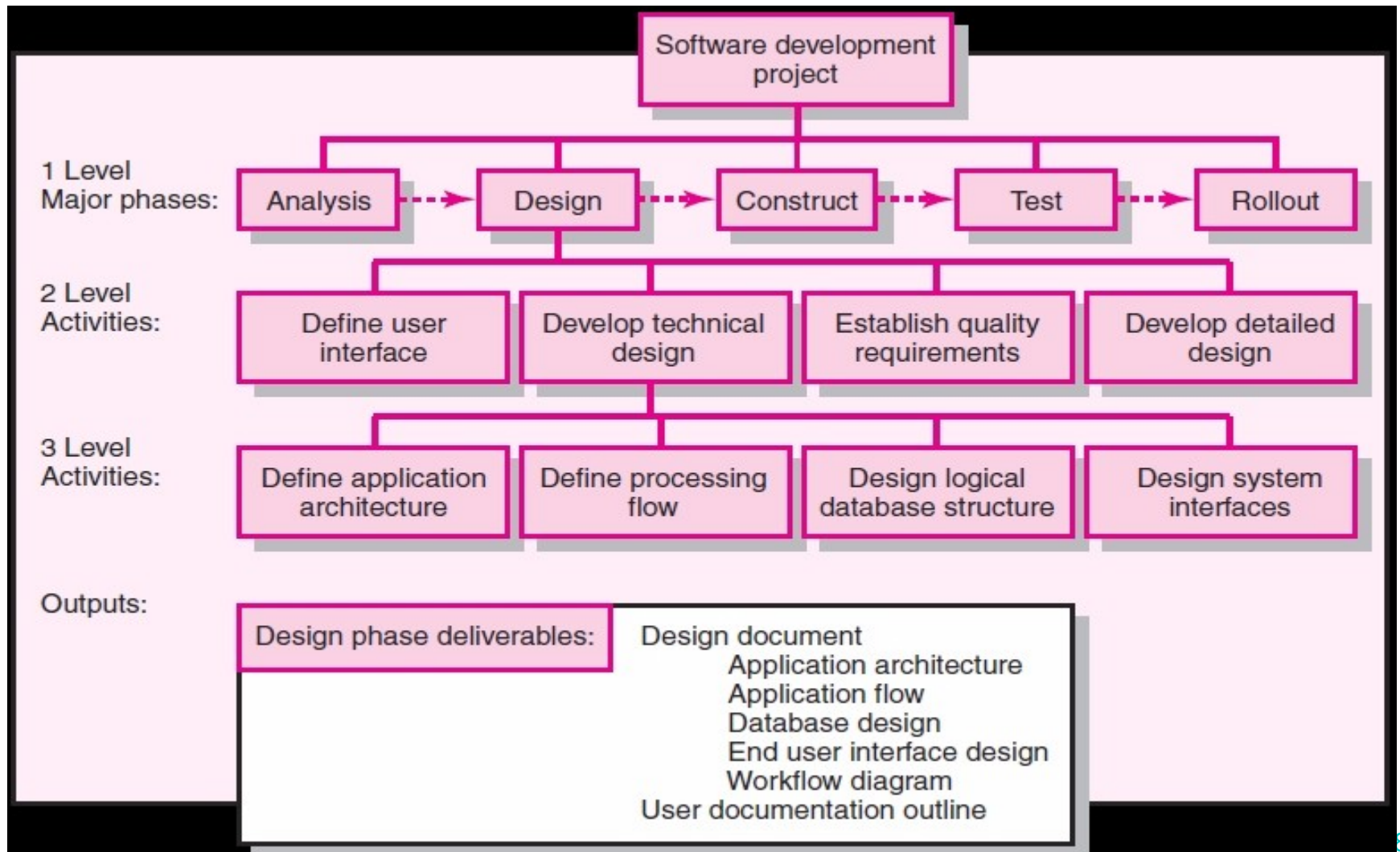
Discussion

- What are the key components of a WBS?
 - Milestones, deliverables, work packages, dependencies, duration, resources, etc
- How can a WBS help in managing project scope and budget?
 - Clear scope, resource planning and allocation, risk identification and planning
- How do you ensure that all project tasks are included in the WBS?
 - Re-visit the scope, break into phases, review and validate with customer, cross-check/reference with other documents

Work Breakdown Structure (WBS)

- A deliverables-oriented grouping of project elements that organizes and defines the total work scope of the project.
- Each descending level represents an increasingly detailed definition of the project work.

Cont'd...Example



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- WBS contains a list of activities for a project derived from
 - » Previous experience
 - » Expert brainstorming
- WBS helps to
 - » Identify the main activities
 - » Break each main activity down into sub-activities which can further be broken down into lower level sub-activities
 - » An instrument for tracking costs and work performance.
 - » Provides a coordinating framework for the various parts of a project.
 - » Defines authority and responsibilities for the details of the project.
 - » Provides the capacity to sum or “roll up” the cost of each project phase.
 - » Identifies “**work packages**”.

Work Package

- Lowest level/unit of WBS.
- Should contain activities that are short in duration (1 or 2 weeks; 80 hour rule; and no longer than a reporting period).
- Work package activities can be completed by an individual or a small team.
- All work packages should be similar in size or effort needed.
 - » [Neither less than 8 hours nor greater than 80 hours]
- Provides input to scheduling and budget development.

Identifying Work Packages

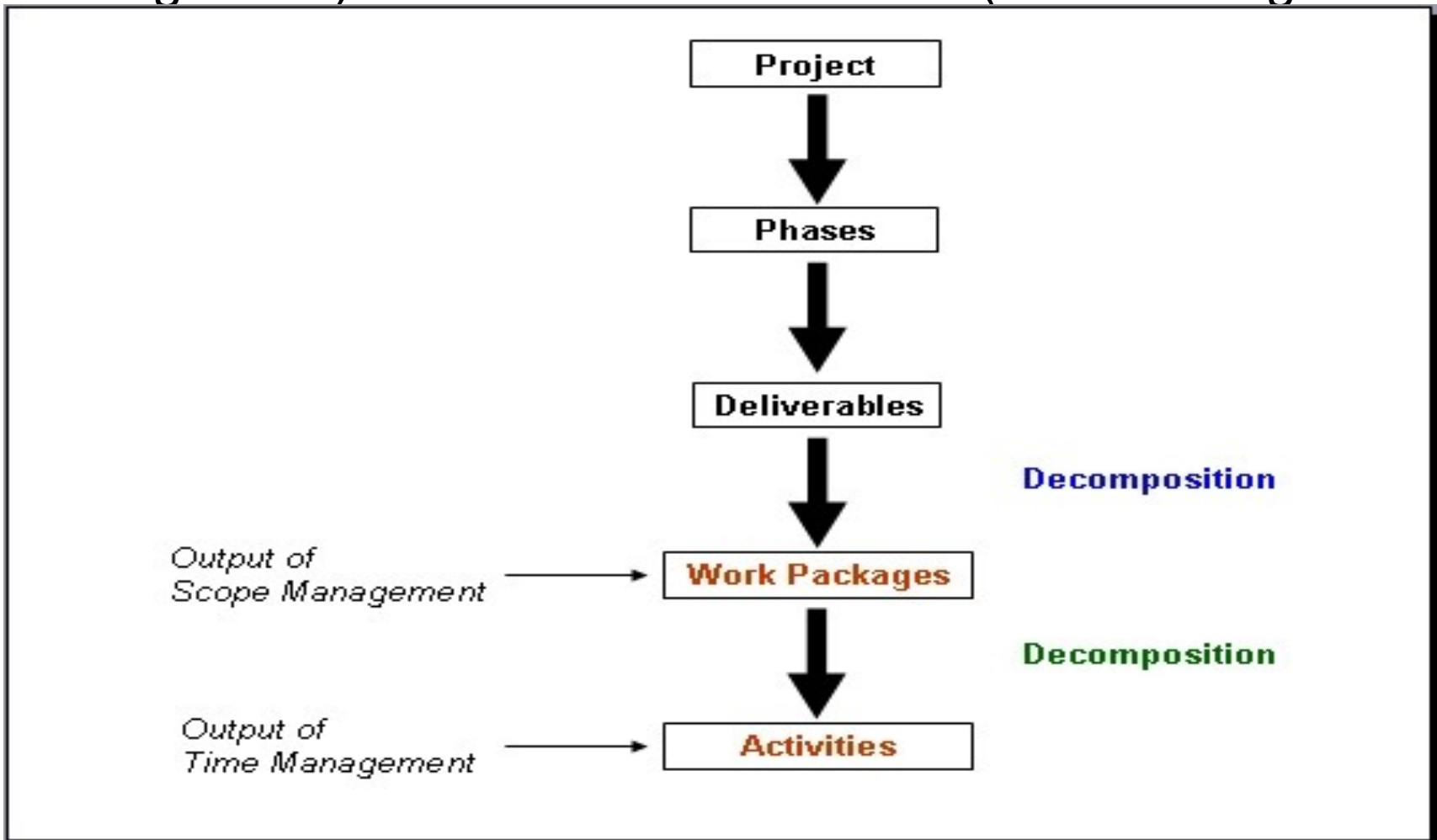
- Identify major project work deliverables/systems.
 - » Then the **sub-deliverables necessary** to accomplish the larger deliverables are defined.
- The process is repeated until the sub-deliverable detail is **small enough to be manageable** and where one person is responsible.
 - » This lowest deliverable usually consists of several work packages.
- Work packages within a deliverable are grouped by type of work: *foundation, framing, finish; hardware, programming, testing, etc.*
- Also referred to as **cost accounts**, work packages facilitate a system for monitoring project progress by **work completed, cost and responsibility**.

WBS Techniques

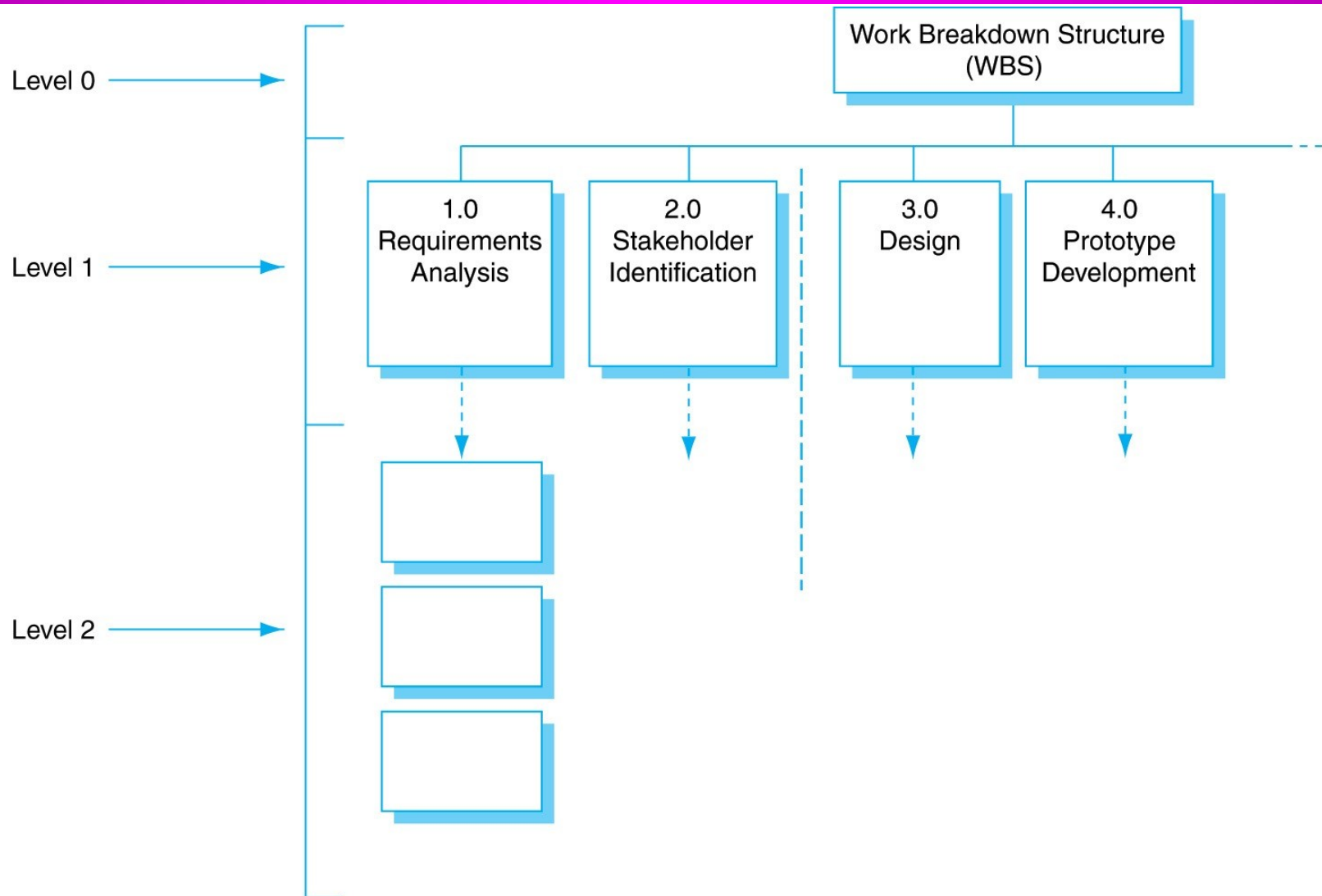
- Decomposition participation includes:
 - » Project team
 - » Customers
 - » Subject matter experts
- Major project deliverables identified
- Codes assigned to each WBS component
 - » Level 0 - project itself
 - » Level 1 - major deliverables
 - » Level 2 - individual components of each deliverable
 - » Etc.
 - » Final level – work package

Project Management on decomposition

- Important technique used in WBS creation (Scope Management) and definition of activities (Time Management).



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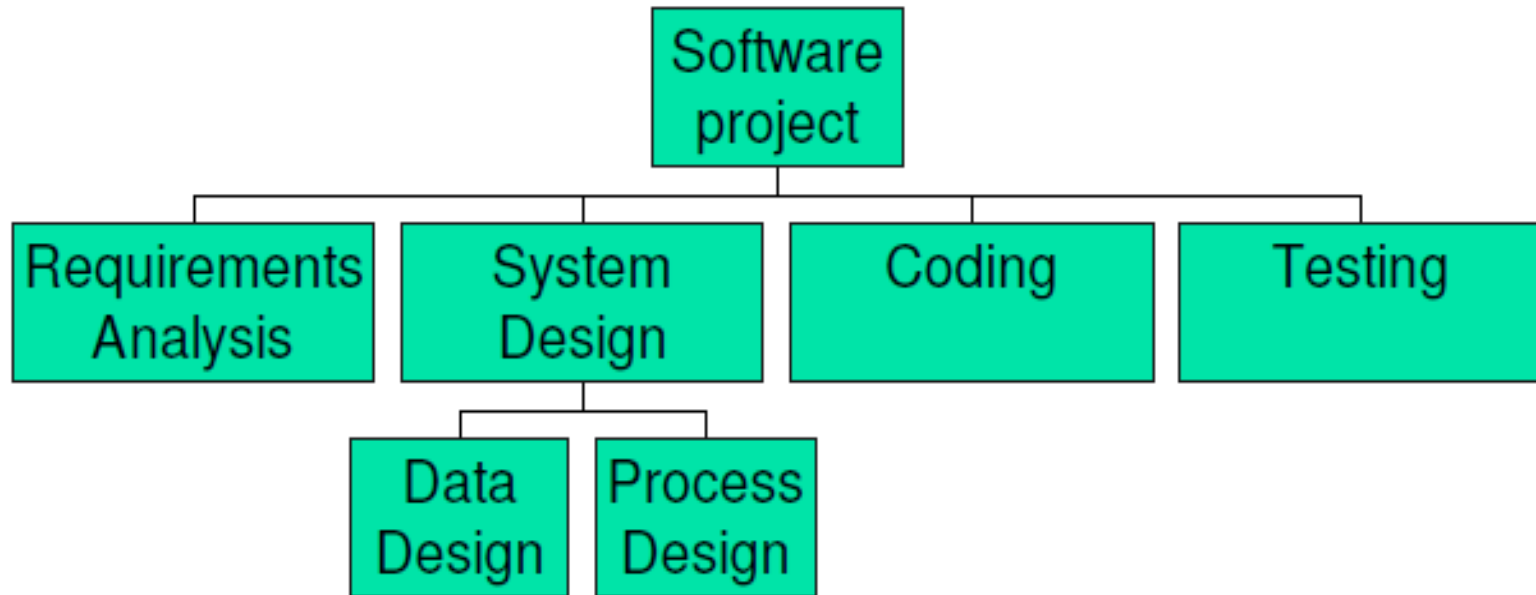
Approaches to Develop WBS

1. Phase based approach
2. Product based approach
3. Hybrid approach

Phase based approach

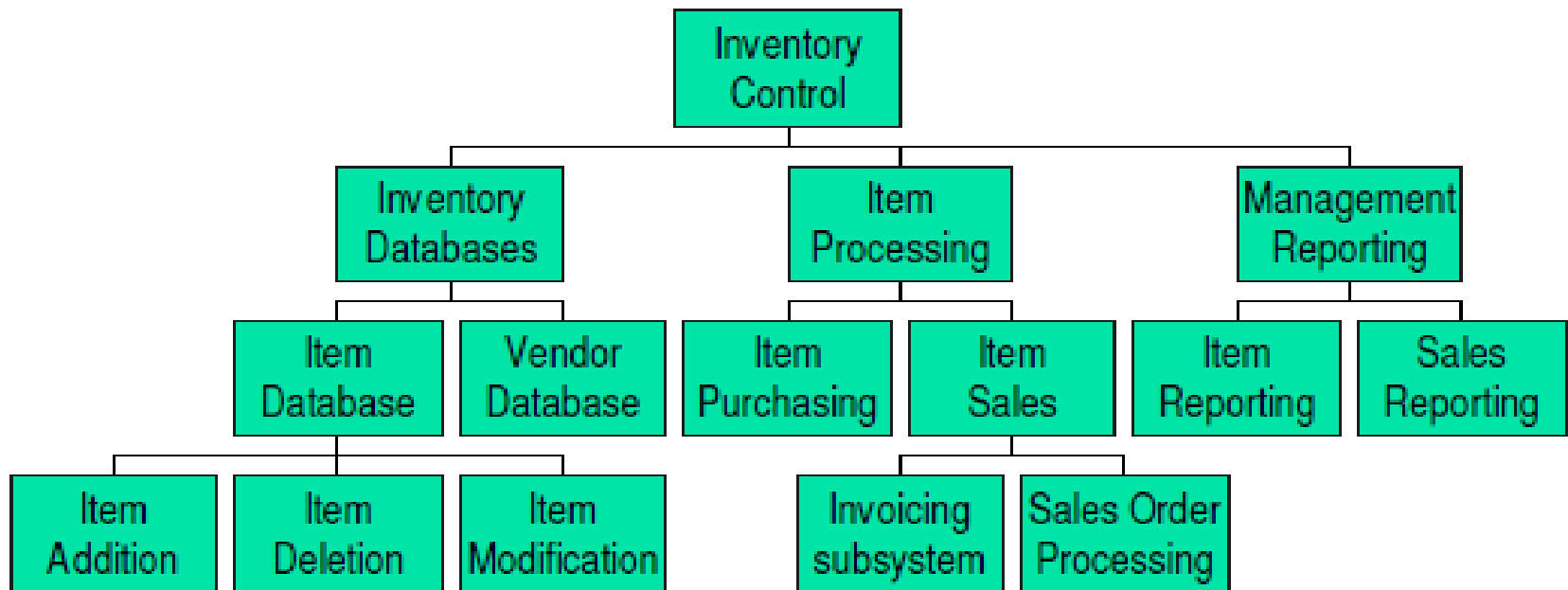
- Advantage
 - » Activity list likely complete and non-overlapping
 - » WBS gives a structure that can be
 - refined as the project proceeds
 - used for determining dependencies among activities
- Disadvantage
 - » May miss some activities related to final product

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Product based approach

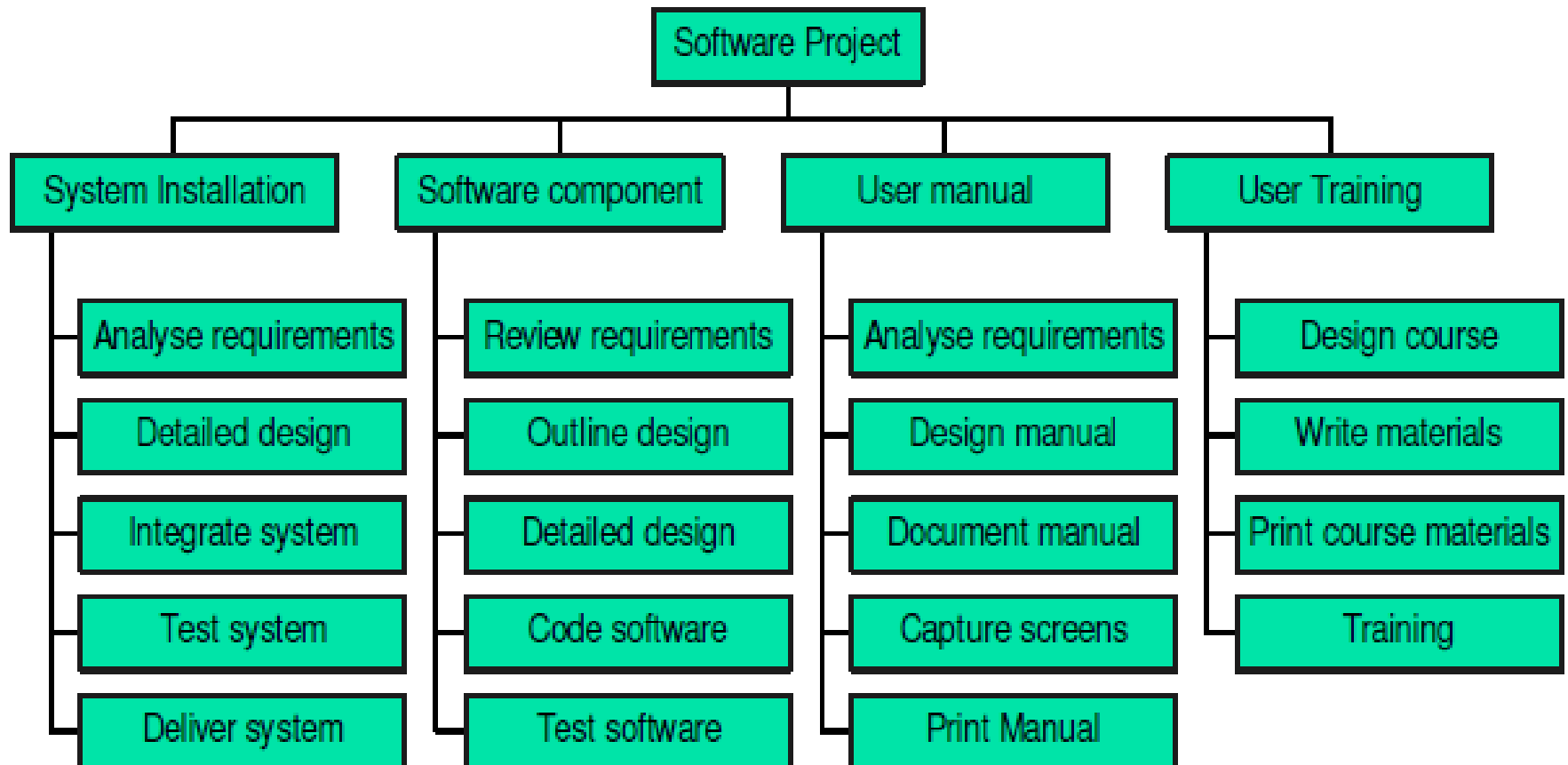
- Product Breakdown Structure (PBS)
 - » Shows how a system can be broken down into different products for development



Hybrid Approach

- A mix of the phase-based and product based approaches (most commonly used)
- The WBS consists of
 - » a list of the products of the project; and
 - » a list of phases for each product

Cont'd... Example



How do you develop a good WBS?

- Top down approach:
 - » Start at the highest, top level activities and systematically develop increasing levels of detail for all activities
- Bottom up approach (“Brainstorming”):
 - » Generate all activities you can think of that will have to be done and then group them into categories
- Which one you use depends on
 - » how familiar you and your team are with the project,
 - » whether similar projects have successfully been performed in the past, and
 - » how many new methods and technologies will be used.

Top Down WBS Development

- Specify all activities for the entire project to be finished.
- Determine all tasks to complete each activity.
- If necessary, specify sub-activities required to complete each task.
- Continue in this way until you have adequately detailed your project.
- **Approach is good if**
 - » You (or your team) are familiar with the problem.
 - » You have successfully managed a similar project in the past.
 - » You are not introducing new methodologies or tools.

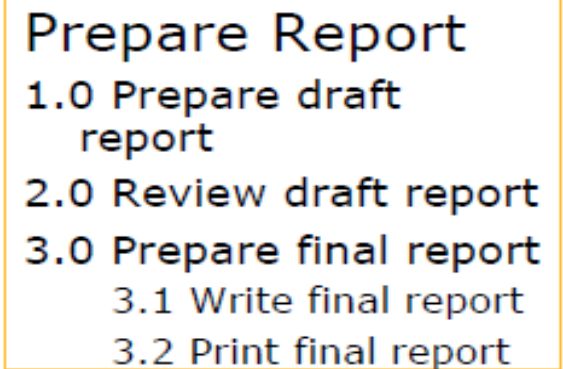
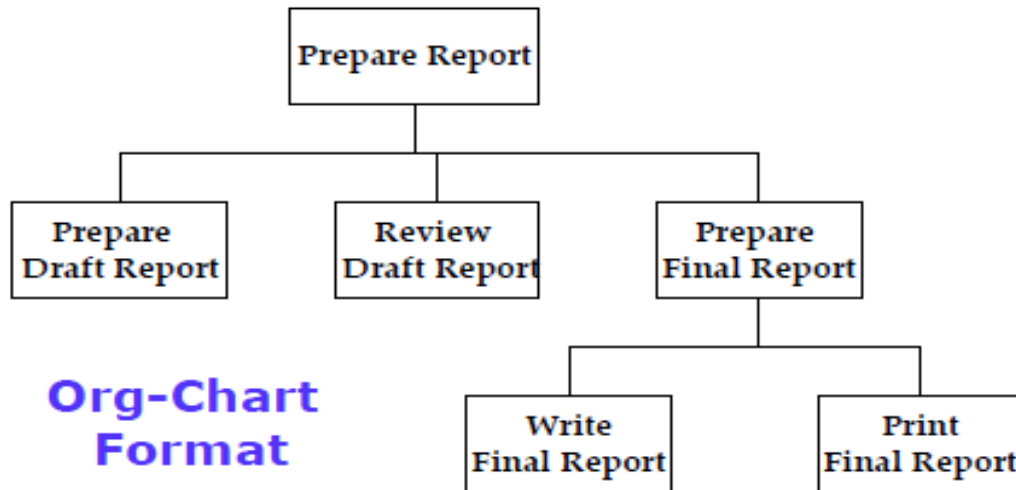
Brainstorming WBS Development

- Brainstorming means you
 - » Don't worry about overlap or level of detail
 - » Don't discuss activity wordings or other details
 - » Don't make any judgements
 - » Write everything down
- On a single list, write any activities you think will have to be performed.
- Then study the list and group activities into a few major categories with common characteristics.
- Try to group activities into higher level activities
- Consider each category you have created
 - » Use *top-down WBS development* to determine any additional activities you may have overlooked.

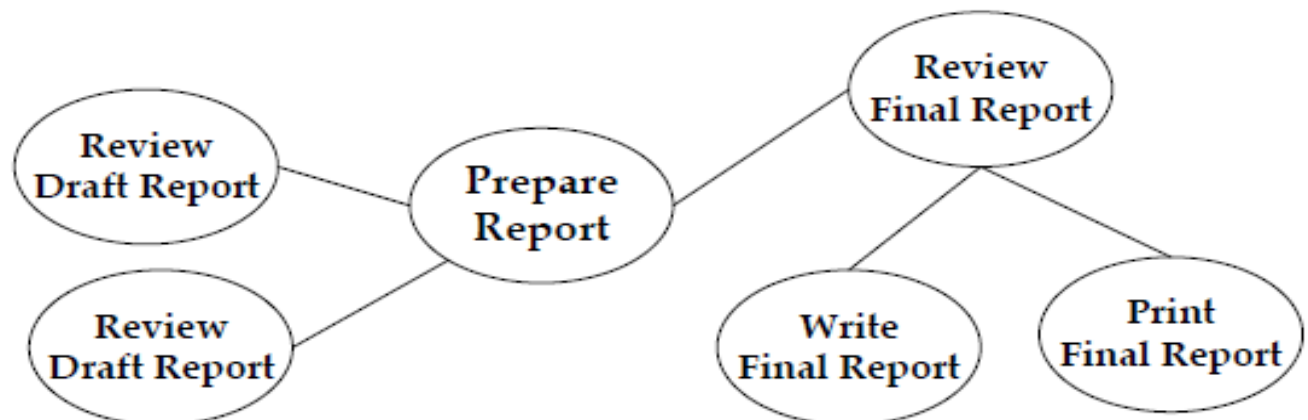
Displaying WBS

- Three different formats are usually used
- **Organization-chart format**
 - » Effectively portrays an overview of your project
 - » Hierarchical relationships of participants, different activities and tasks
- **Outline format**
 - » Sub-activities and tasks are indented
- **Bubble format**
 - » The bubble in the center represents the project
 - » Lines from the center bubble lead to activities
 - » Lines from activities lead to tasks.

Cont'd



Bubble Format



Cont'd

- **What is the best display format for WBS?**
 - » Organization-chart format:
 - Often good for a “bird view” (executive summaries,...)
 - Less effective for displaying large numbers of activities
 - » Outline format:
 - Easier to understand, if WBS contains many activities
 - » Bubble format:
 - Effective for supporting brainstorming
 - Not so good for displaying work breakdown structures to audiences who are not familiar with the project.
 - » Mixed approach
 - In large projects

Tips for Breaking Down the Project

- Breakdown the work until you can do an estimate that is accurate enough for your purposes.
 - » If you are pricing a project to submit a competitive bid then you are likely to go down to the work package level.
- Tasks should be limited as best as possible to one day or more to complete.
- Activities should have clearly defined start and end
 - » Avoid open-ended tasks like “research” or “market analysis.”
- Accountability and control are important! Break the work down so that one individual is clearly responsible for each unit.

Activity Networks and Bar Charts

- Graphical notations are often used to illustrate project schedules.
- **Activity charts** (a.k.a. *PERT - Program Evaluation and Review Technique* charts) show task dependencies, durations, and the *critical path*.
- **Bar charts** (a.k.a. *GANTT* charts) generally show resource (e.g., people) assignments and calendar time.

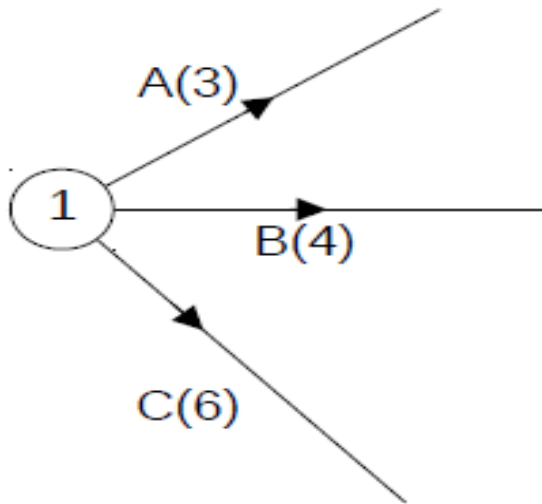
Activity Network

- The table below shows the tasks involved in a project, with their durations and immediate predecessors:

Tasks	Duration (hours)	Immediate Predecessors
A	3	-
B	4	-
C	6	-
D	5	A
E	1	B
F	6	B
G	7	C, D, E

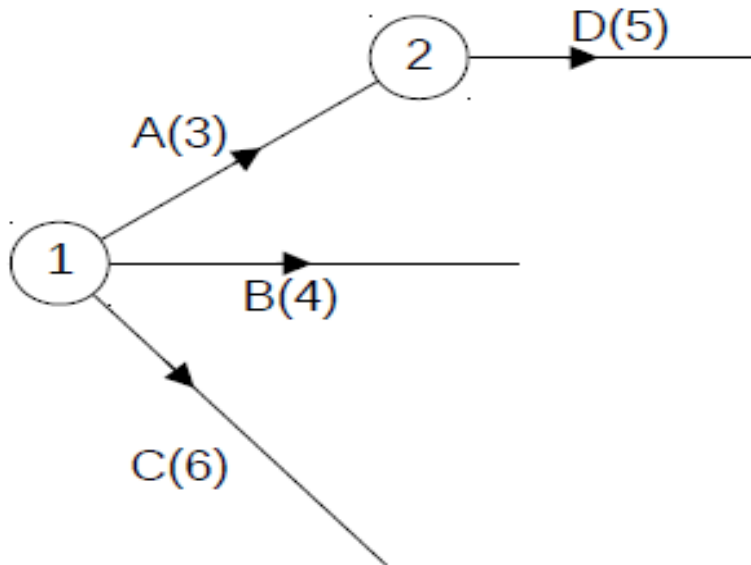
- Draw an activity network for the project.

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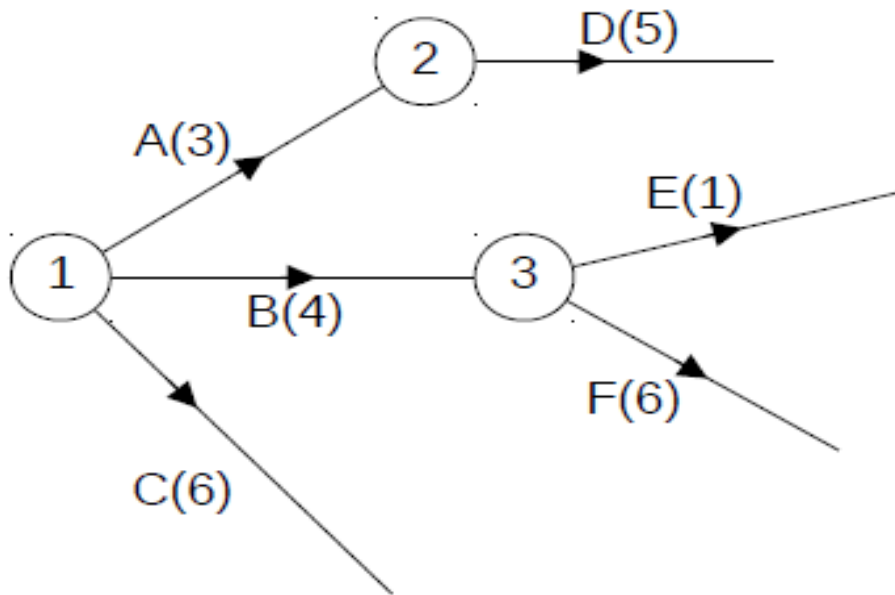
- First draw a start node labelled 1.
- Activities A, B and C do not depend on any other activity, so they all begin at node 1.

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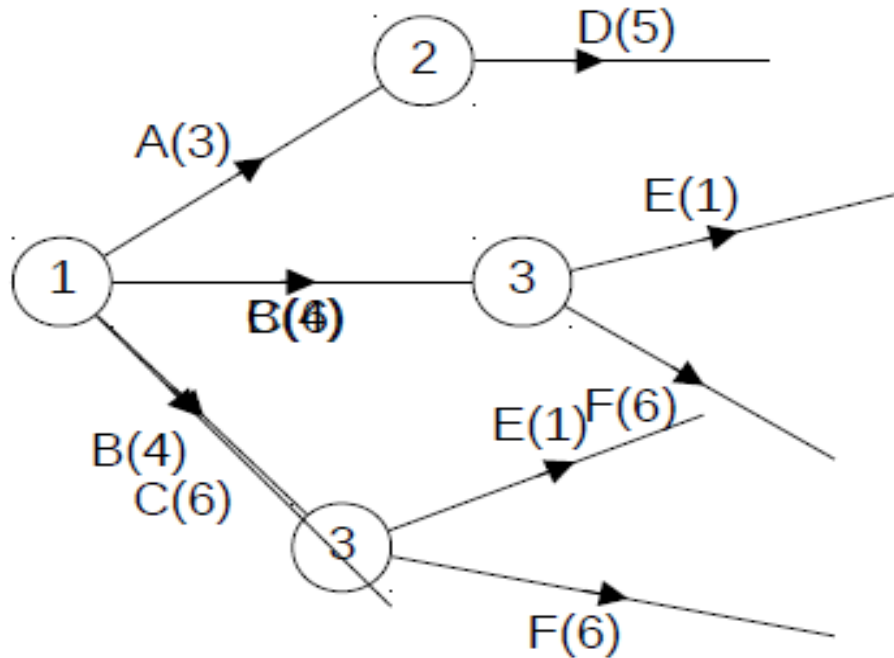
- Activity D depends on A, so add event node 2 at the end of A.
- Then add activity D.

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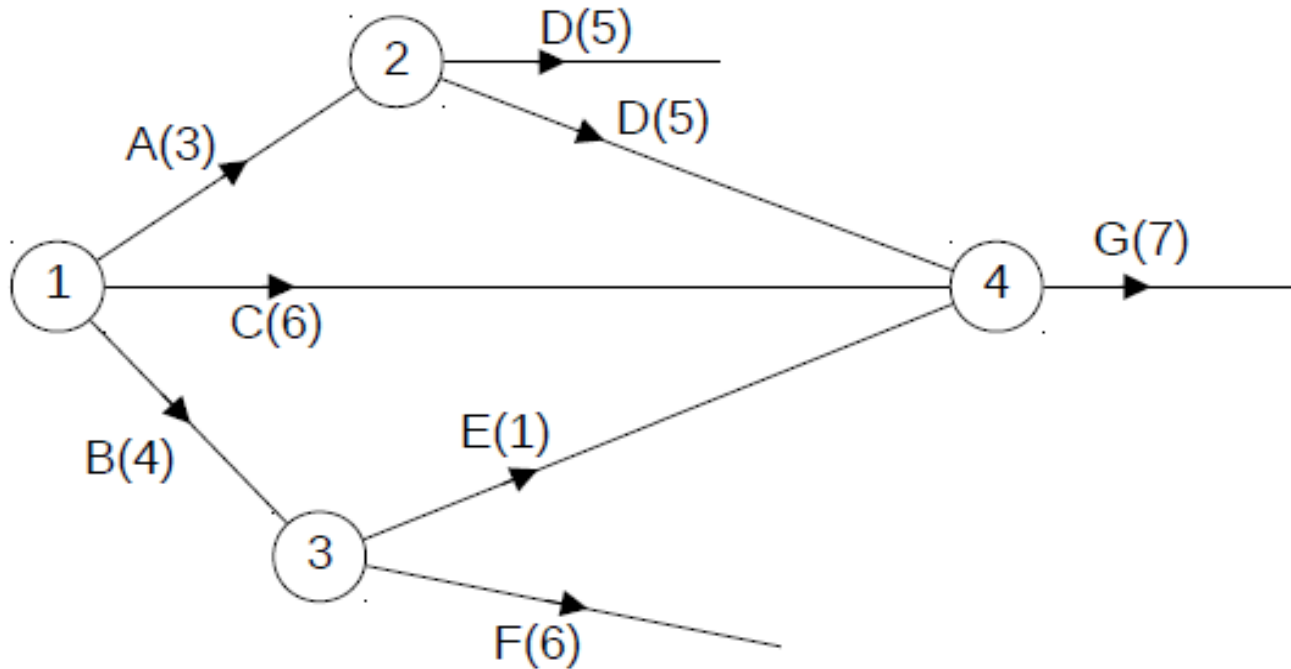
- Activities E and F both depend on B, so add event node 3 at the end of B.
- Then add activities E and F.

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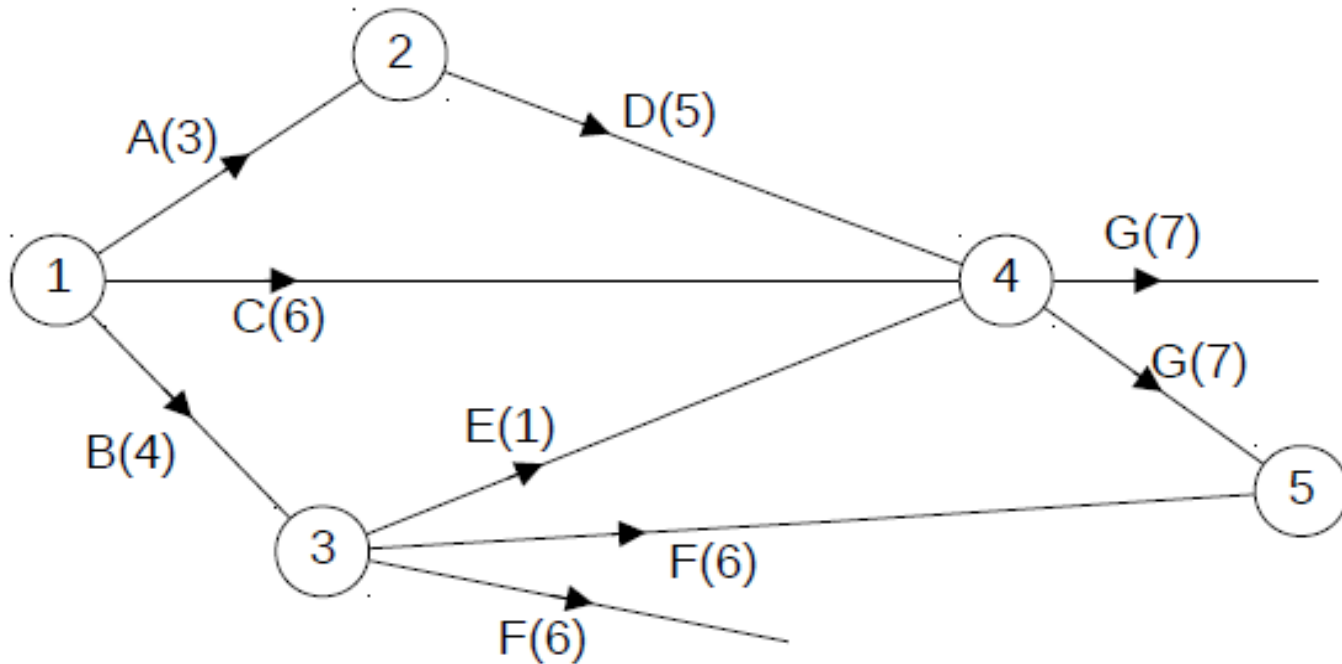
- Activity G depends on C, D and E, so all these three events need to end at the same node.
- This is easiest if you redraw the network so that C is between A and B.

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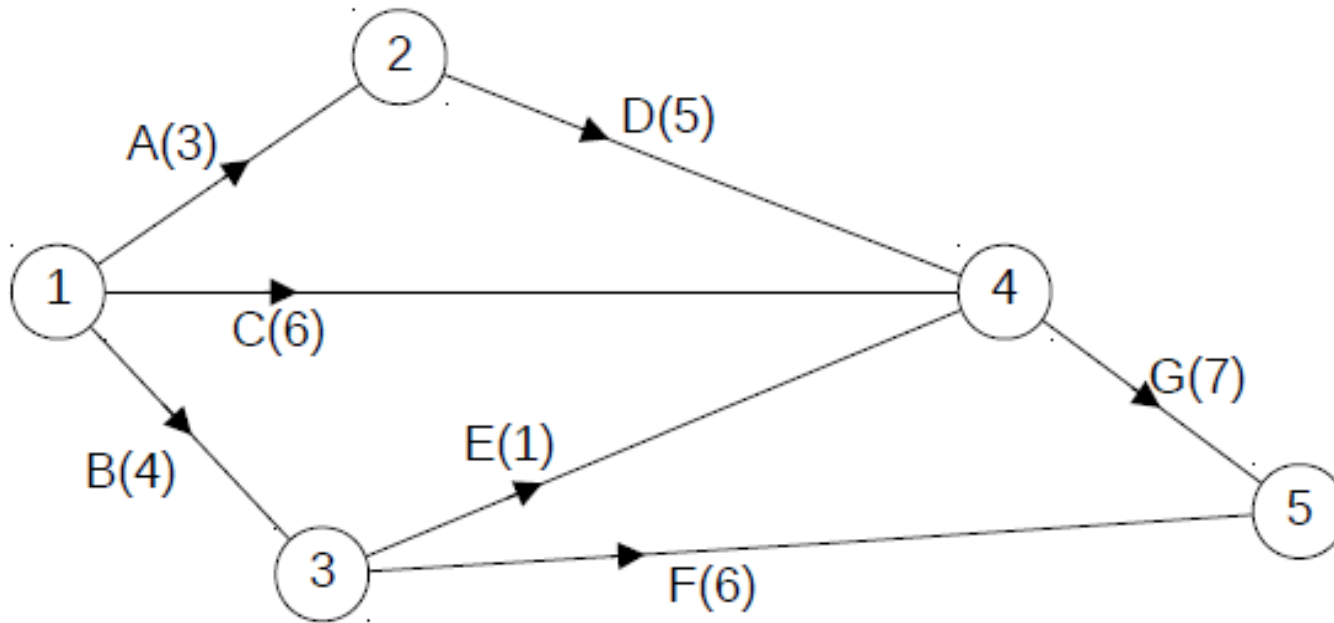
- Now add node 4, with C, D and E leading into it.
- Then add activity G.

Cont'd



- A finish node is now needed.
- Any activities not leading into a node must end at the finish node.

Cont'd



- The network is now complete.

Exercise

- Draw an activity network for the following table:

Task	Duration(days)	Dependencies
T1	8	
T2	15	
T3	15	T1(M)
T4	10	
T5	10	T2, T4(M)
T6	5	T1, T2(M)
T7	20	T1(M)
T8	25	T4(M)
T9	15	T3, T6(M)
T10	15	T5, T7(M)
T11	7	T9(M)
T12	10	T11(M)

PERT – Program Evaluation and Review Technique

- Developed by the US Navy for the planning and control of the Polaris missile program
- Developed for R&D projects where activity times are generally uncertain
- Primary objectives:
 - » Shortest possible time
 - » Coping with uncertain activity completion times, e.g.:
 - For a particular activity the most likely completion time is 4 weeks but it could be anywhere between 3 weeks and 8 weeks

CPM – Critical Path Method

- Primary objectives:
 - » Plan for the fastest completion of the project
 - » Identify activities whose delays is likely to affect the completion date for the whole project
 - » Very useful for repetitive activities with well known completion time
- Developed by Du Pont Chemical Company and published in 1958
 - » Can we decrease the completion time by spending more money

CPM Calculation

- The forward pass
 - » calculate the **earliest** start dates of the activities
 - to calculate the project completion date
- The backward pass
 - » calculate the **latest** start dates for activities
 - to identify the critical path from the graph

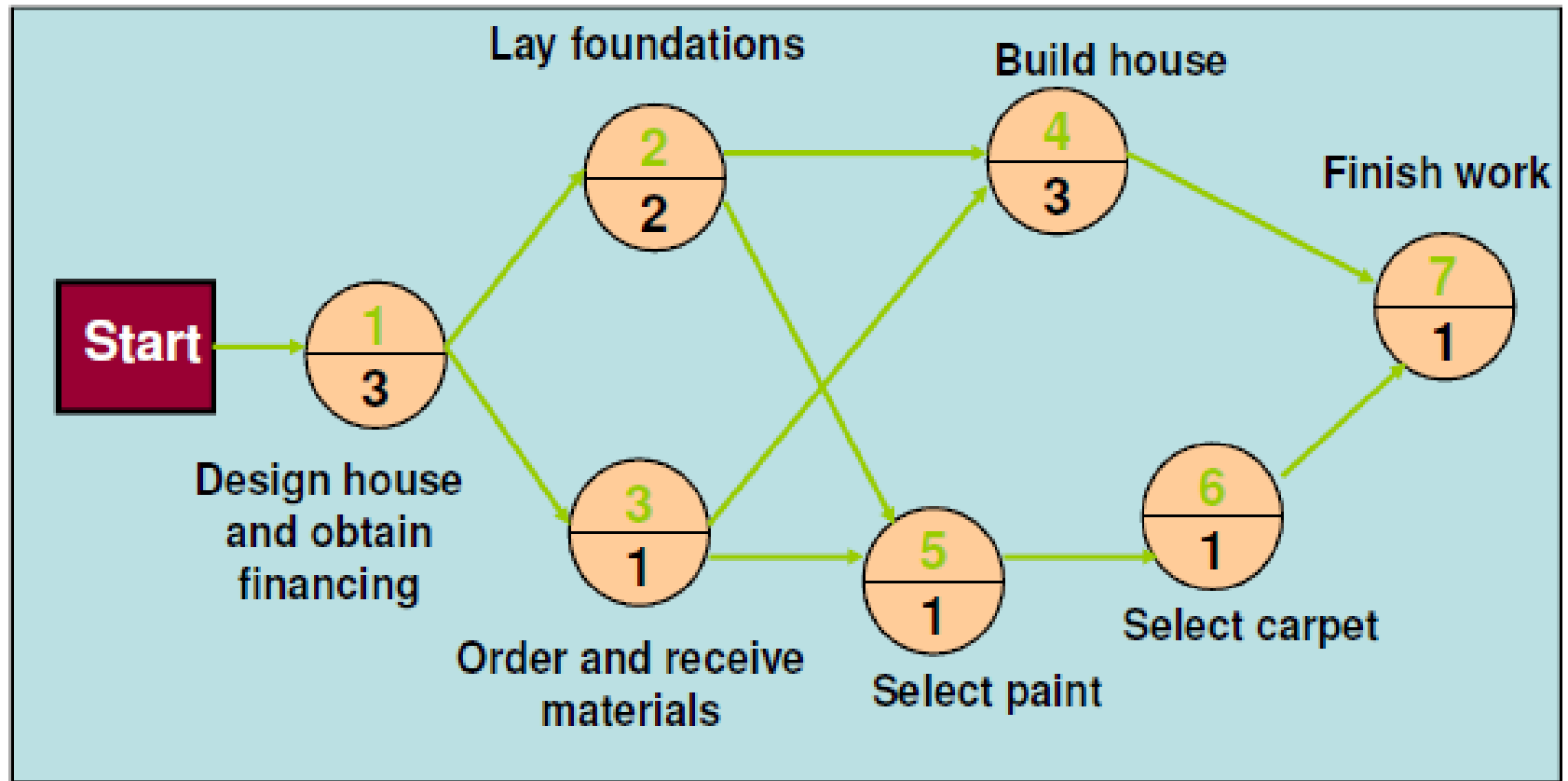
Critical Path and Events

- **Critical event:** an event that has zero ***slack***
- **Critical path:** a path joining those critical events
- **Benefit of Critical Path Analysis:**
 - » During planning stage
 - Shortening the critical path will reduce the overall project duration
 - » During management stage
 - Pay more attention to those activities which fall in the critical path
- **Float (slack)** - amount of time that a task can be delayed without causing a delay to:
 - » subsequent tasks (free float)
 - » project completion date (total float)

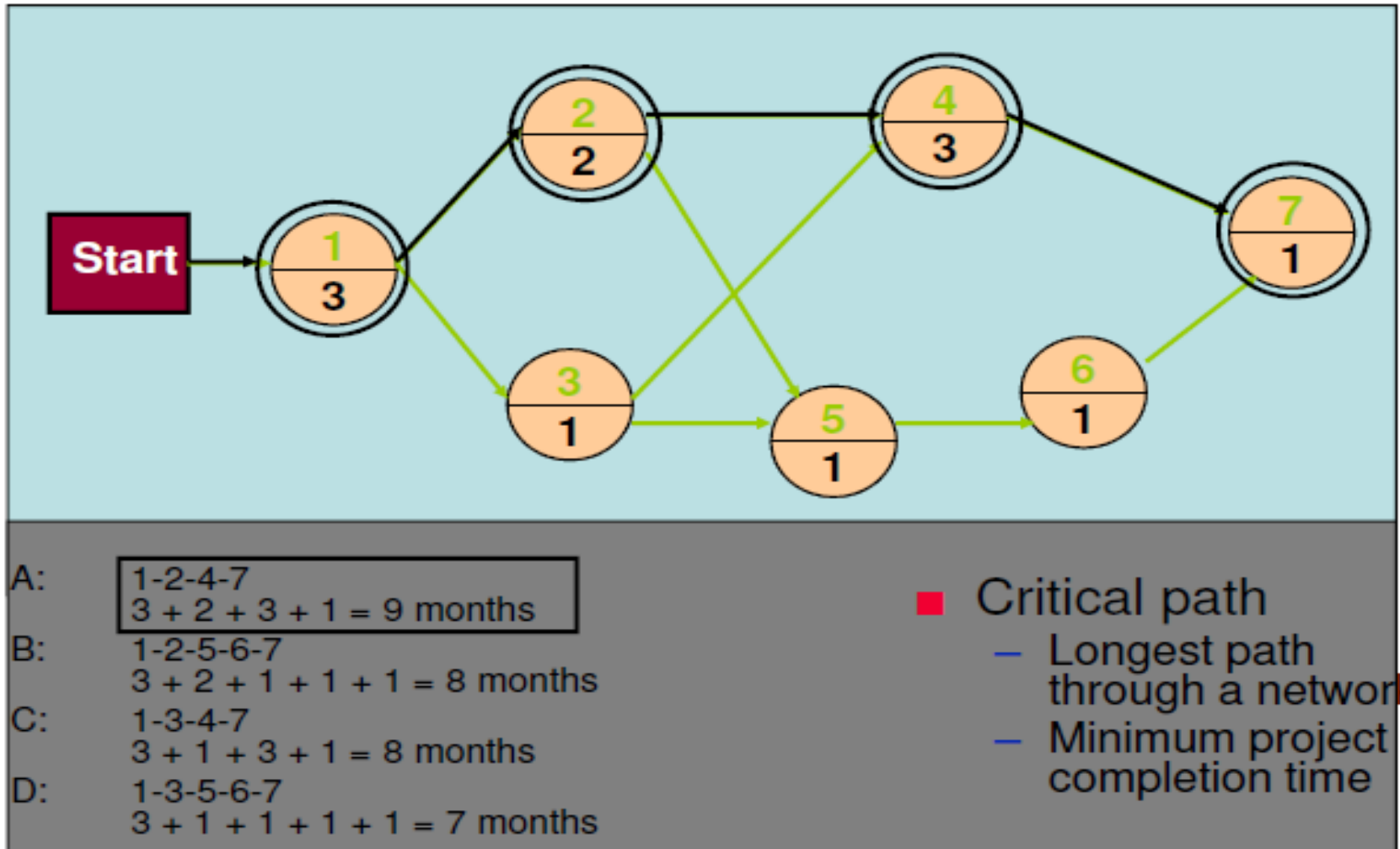
Activity Float

- Time allowed for an activity to delay
- Three different types:
 - » **Total float** (without affecting project completion)
= latest start date – earliest start date
 - » **Free float** (without affecting the next activity)
= earliest start date of *next* activity – latest end date of *previous* activity
 - » **Interfering float** (= total float - free float)

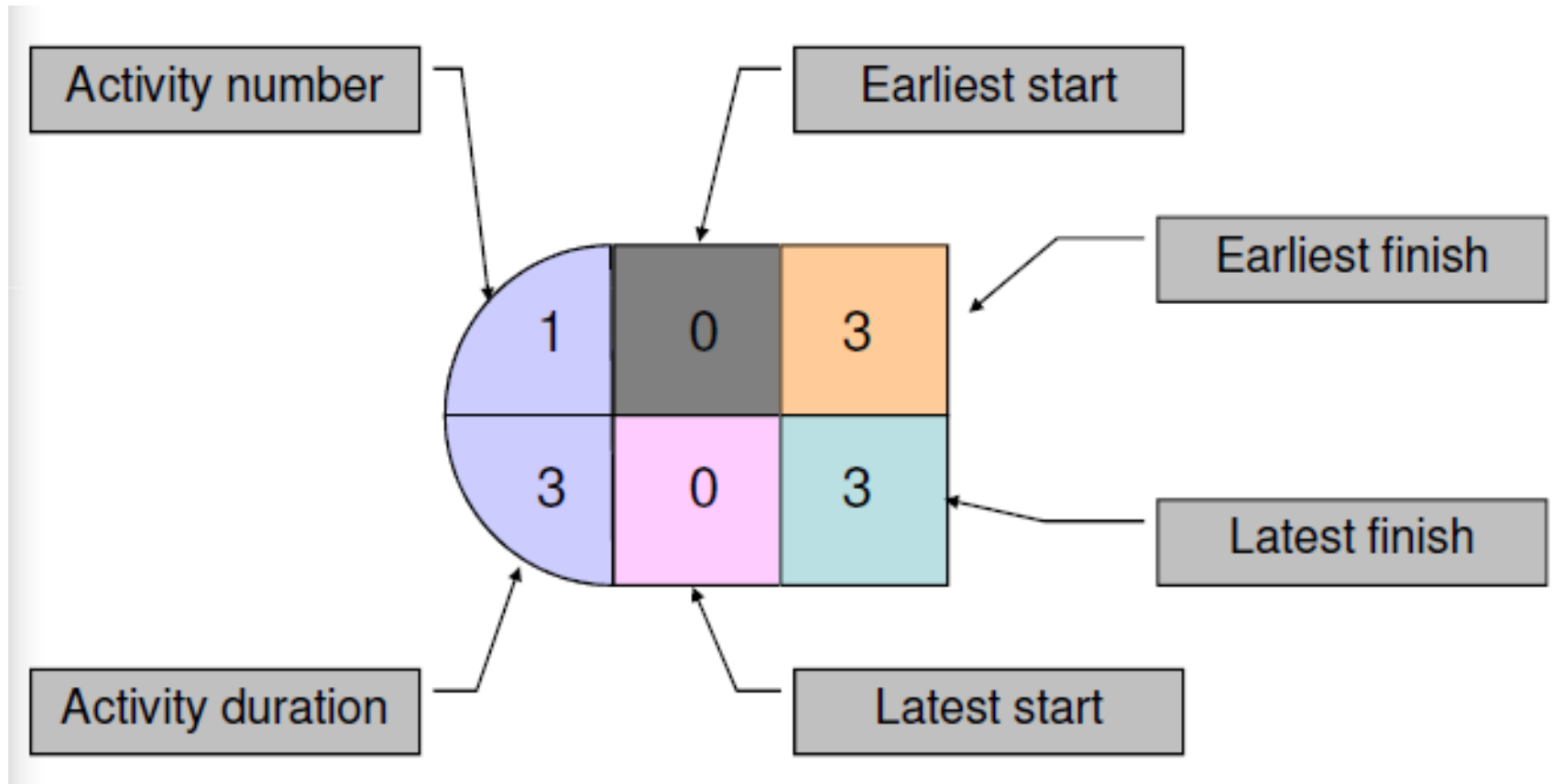
Example - Network for House Building Project



Critical Path



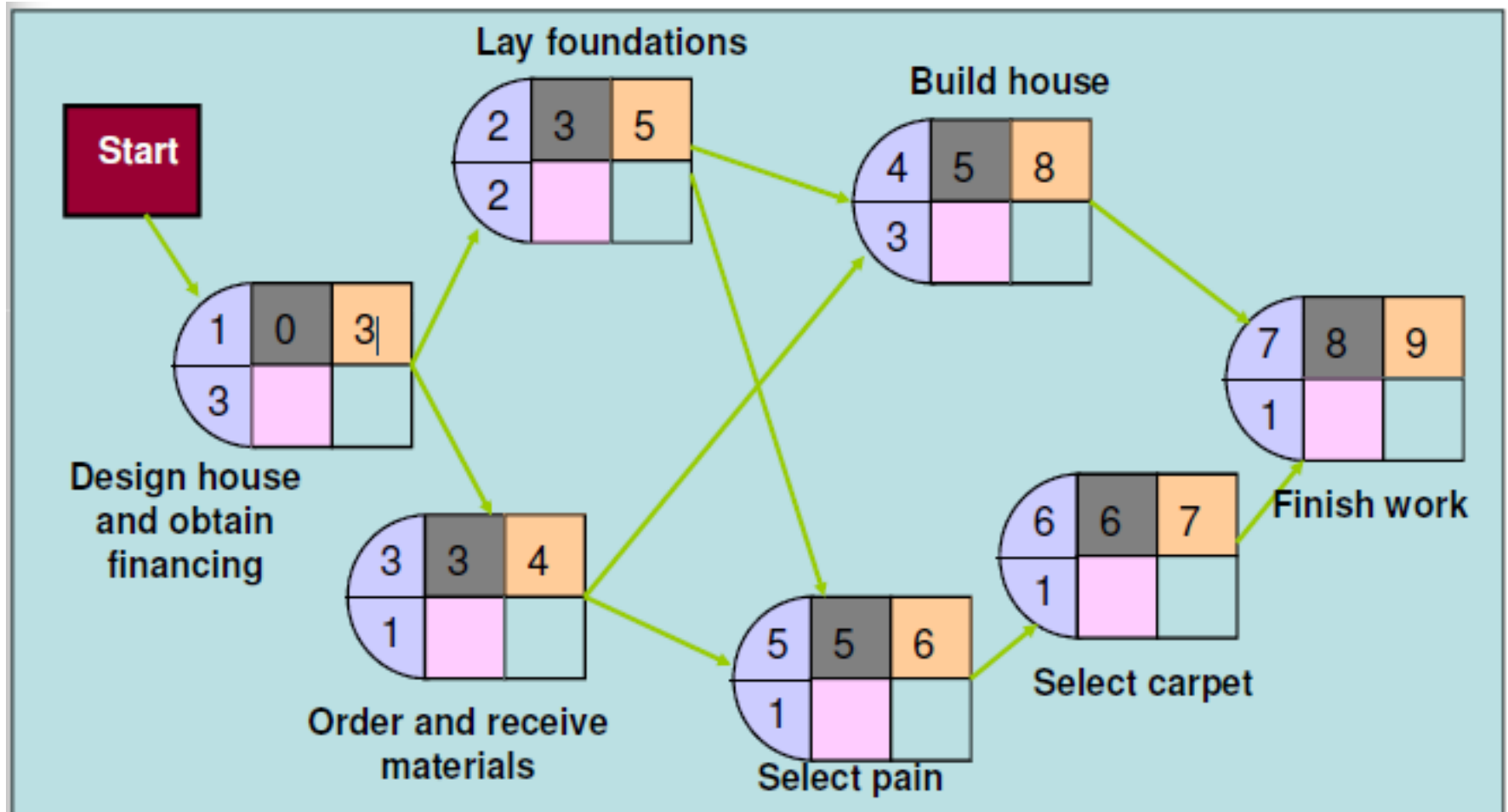
Mode of Configuration (Activity Identity Box)



Forward Pass

- Start at the beginning of CPM/PERT network to determine the earliest activity times
- Earliest Start Time (ES)
 - » earliest time an activity can start
 - » $ES = \text{maximum EF of immediate predecessors}$
- Earliest finish time (EF)
 - » earliest time an activity can finish
 - » earliest start time plus activity time
 - » $EF = ES + t$

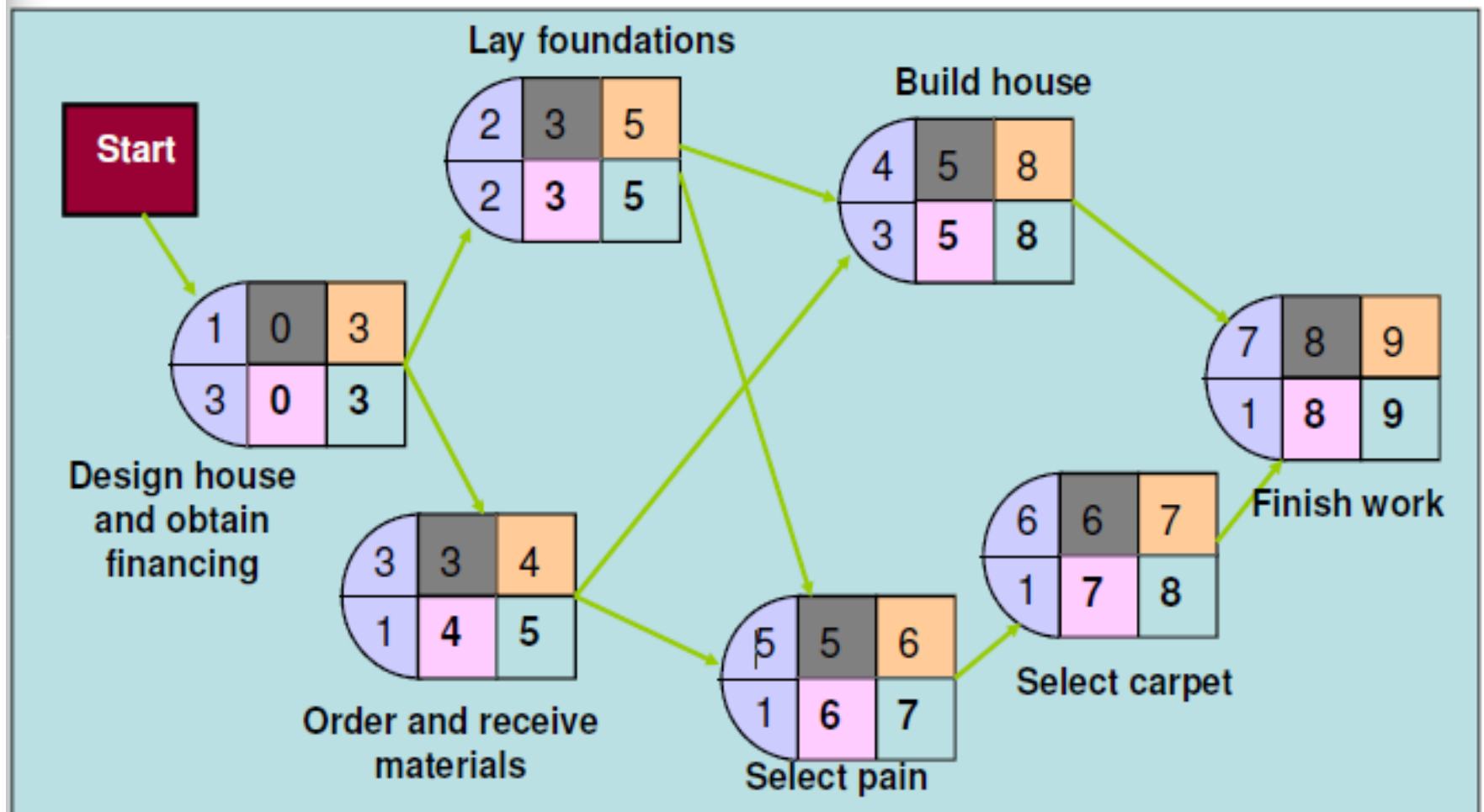
Earliest Activity Start and Finish Times



Backward Pass

- Determines latest activity times by starting at the end of CPM/PERT network and working forward
- Latest Start Time (LS)
 - » Latest time an activity can start without delaying critical path time
 - » $LS = LF - t$
- Latest finish time (LF)
 - » latest time an activity can be completed without delaying critical path time
 - » $LS = \text{minimum } LS \text{ of immediate predecessors}$

Latest Activity Start and Finish Times



Activity Slack

Activity	LS	ES	LF	EF	Slack S
*1	0	0	3	3	0
*2	3	3	5	5	0
3	4	3	5	4	1
*4	5	5	8	8	0
5	6	5	7	6	1
6	7	6	8	7	1
*7	8	8	9	9	0
* Critical Path					

- **Slack:** amount of time an activity can be delayed without delaying the project
- **activity slack** = $LS - ES = LF - EF$
- **Critical activities:** have zero slack and lie on a critical path.

GANTT Charts

- Named after Henry Gantt (1861 - 1919) an American mathematical engineer.
- Graph or bar chart with a bar for each project activity that shows passage of time
- A Gantt chart is a type of bar chart that illustrates a project schedule.
- Gantt charts illustrate the start and finish dates of the work elements (tasks) of a project.
- Elements of a GANTT chart:
 - » Task names
 - » Start and finish dates of each tasks (graphically)
 - » Dependency relationships
 - » Task duration in an additional column
 - » Lag relationships (start-to-start, finish-to-start etc.)
 - » Name of the project worker responsible for the task or
 - » Resource specifications

GANTT chart example

ID	Activities	Who is responsible ?	Time intervals							
			1.	2.	3.	4.	5.	6.	7.	8.
1.										
2.										
3.										
4.										
5.										
6.										
7.										
8.										
9.										
10.										
11.										

Benefits of the GANTT chart

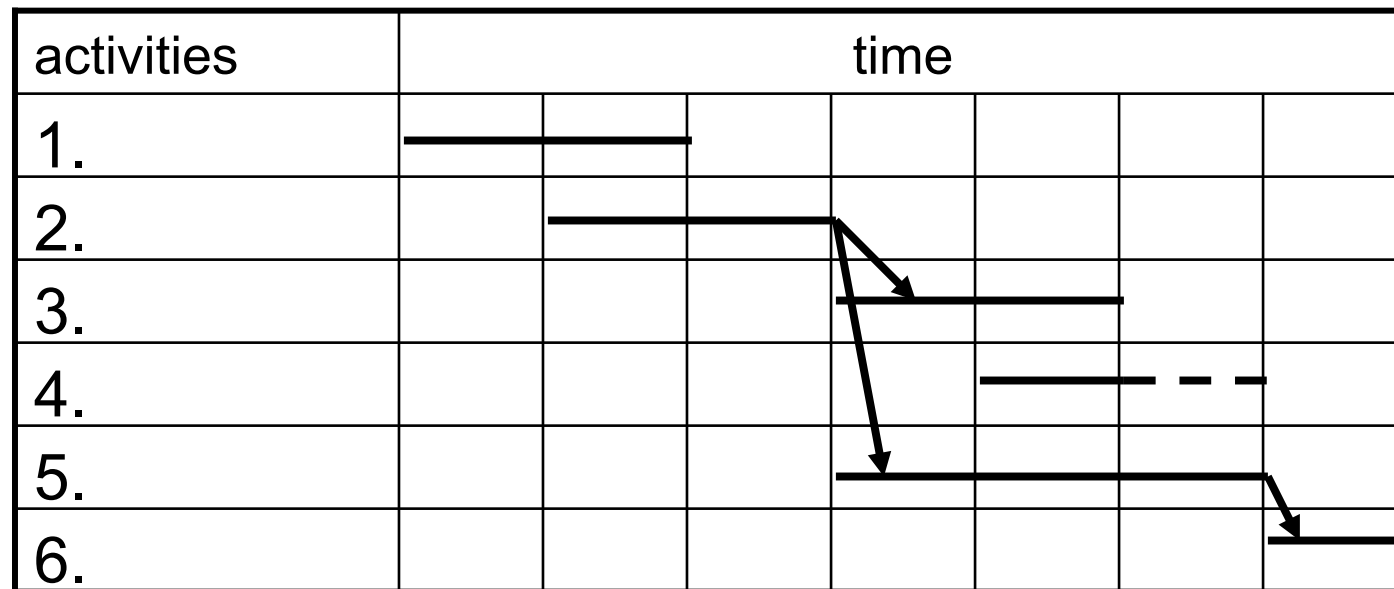
- Easy to read and comprehend
- Easy to create
- Identify the project network coupled with its schedule baseline
- Allow for updating and project control
- Useful for resource planning

Limitations

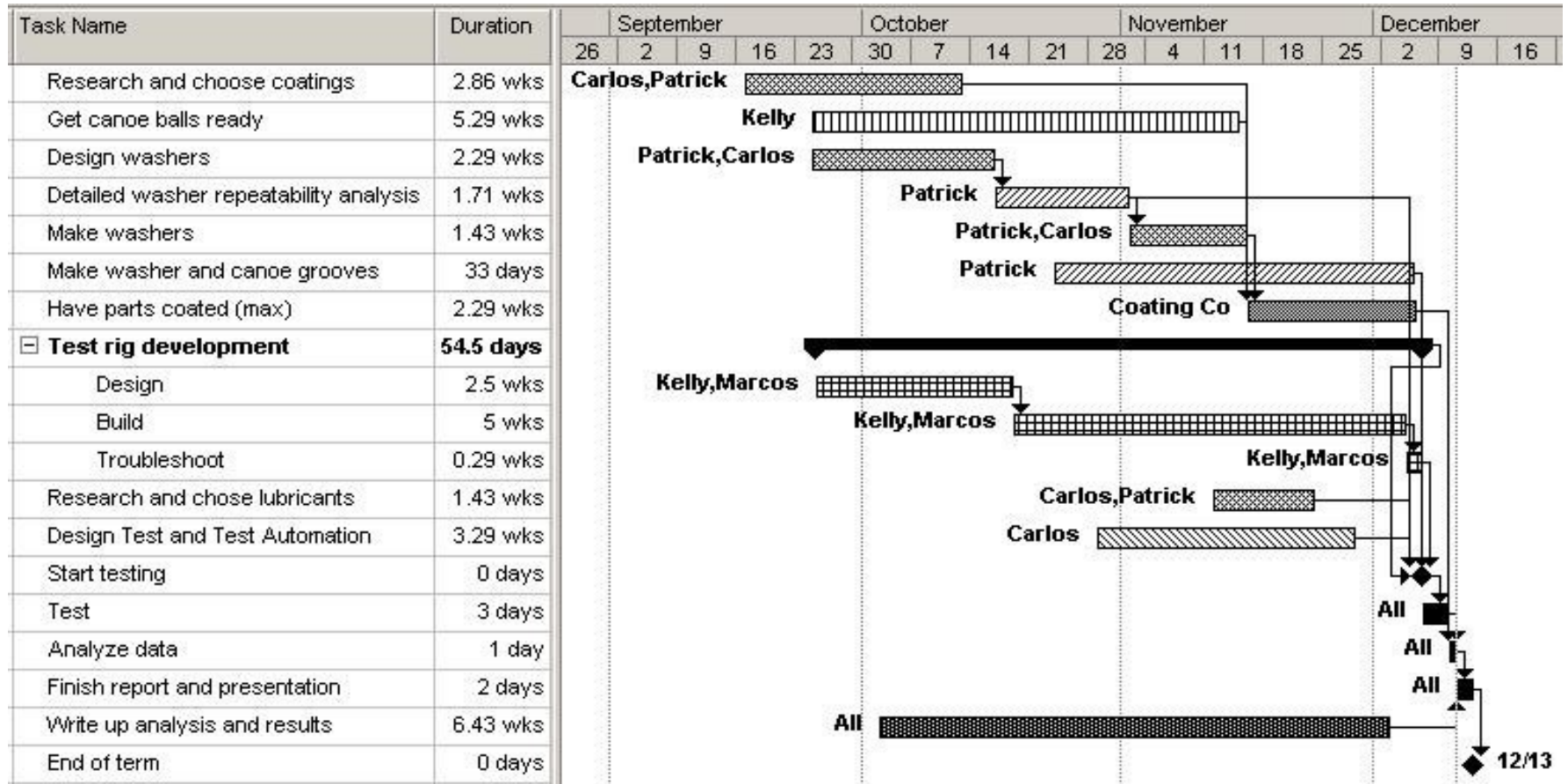
- Can become quite unwieldy for projects with too many activities.
- Projects are often too complex for a Gantt chart.
- Gantt charts represent only a part of the project constraints.
- The technique of creating a GANTT depends on the features of the project.

GANTT chart example

Some Gantt charts also show the dependency relationships between activities.



GANTT chart example



GANTT chart example

Gantt charts can be used to show current schedule status using percent-complete shadings, or colour lines and a vertical "TODAY" line.

